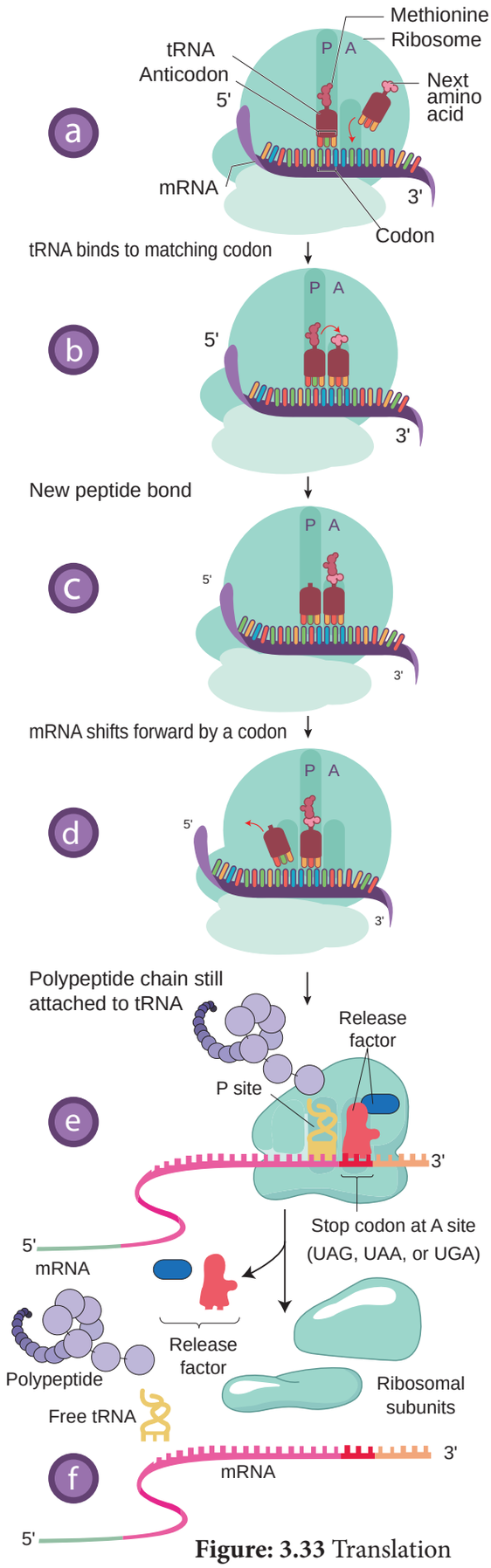


Figure: 3.33 Translation

tRNA is held and linked to the growing end of the polypeptide chain.

ii. **A-Site – The Aminoacyl – tRNA binding site**. This is another tRNA binding site which holds the incoming amino acids called aminoacyl tRNA. The anticodons of tRNA pair with the codons of the mRNA in these sites.

2. Elongation of polypeptide chain

The P and A sites are nearby, so that two tRNA form base pairs with adjacent codon. The polypeptide chain is formed by the pairing of codons and anticodons according to the nucleotide sequence of the mRNA.

Translators of the genetic code - tRNA

The tRNA translates the genetic code from the nucleic acid sequence to the amino acid sequence i.e from gene – Polypeptide. When an amino acid is attached to tRNA it is called **aminoacylated or charged.** This is an energy requiring process which uses the ATP for its energy requirement. Protein synthesis takes place as the next aminoacyl tRNA binds to the A-Site.

The translation begins with the AUG codon (start codon) of mRNA. The tRNA which carries first amino acid **methionine** attach itself to P-site of ribosome. The ribosome adds new amino acids to the growing polypeptides. The second tRNA molecules has anticodons which carries amino acid **alanine** pairs with the mRNA codon in the A-site of the ribosome. The aminoacids **methionine** and **alanine** are close enough so that a peptide bond is formed between them.

The bond between the first tRNA and **methionine** now breaks. The first tRNA leaves the ribosome and the P-site is vacant. The ribosome now moves one codon along the mRNA strand. The second t-RNA molecule now occupies the P-site. The third t-RNA comes and fills the A site (serine). Now a peptide bond is formed between **alanine** and **serine.** The mRNA then moves through the

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